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Vascular System

Vascular tissues are in **radial arrangement**. The tissue by which xylem and phloem are separated is called **conjunctive tissue**. In bean, the conjuctive tissue is composed of parenchyma tissue. Xylem is in **exarch condition**. The number of protoxylem points is four and so the xylem is called **tetrach**. Each phloem patch consists of sieve tubes, companion cells and phloem parenchyma. Metaxylem vessels are generally polygonal in shape. But in monocot roots they are circular.

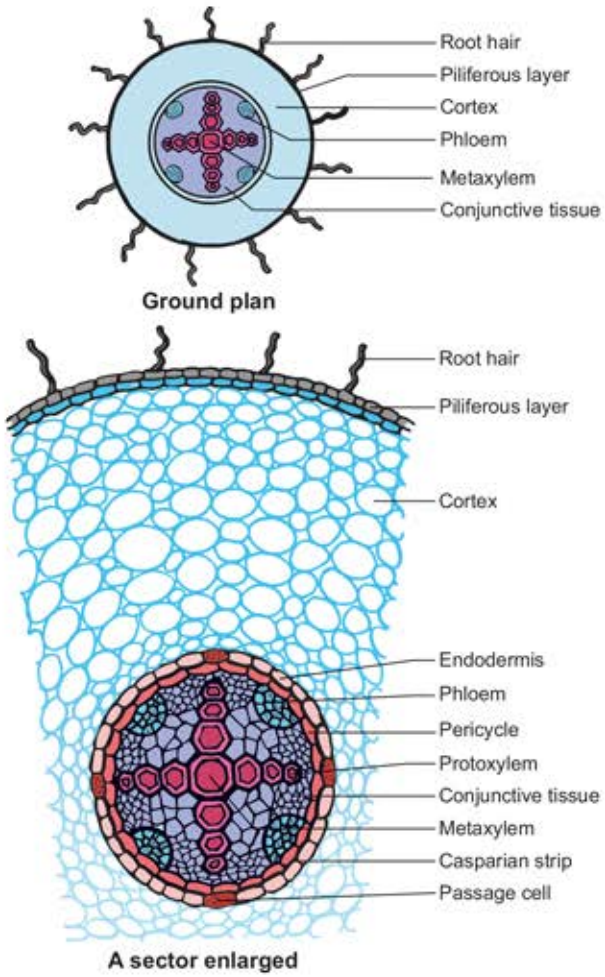


Figure 9.19: T.S. of Dicot root (Bean root)

Primary Structure of Monocot Root-maize Root

The transverse section of the monocot root (maize) shows the following plan of arrangement of tissues from the periphery to the centre.

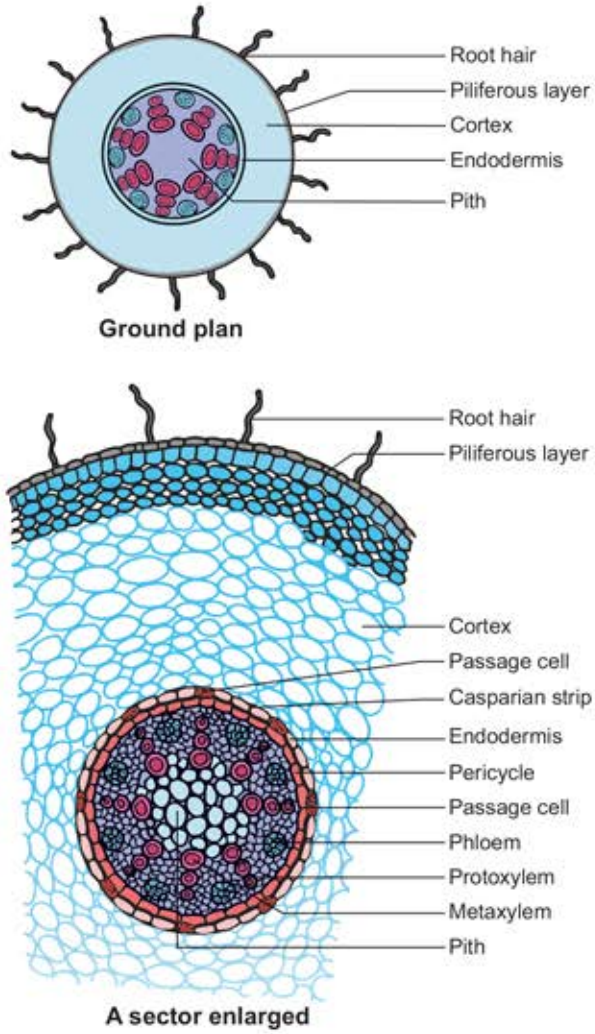


Figure 9.20: T.S of Monocot root (Maize root)

Piliferous Layer or Epiblema

The outermost layer of the root is known as **piliferous layer**. It consists of a single row of thin-walled parenchymatous cells without any intercellular space. Epidermal pores and cuticle are absent in the piliferous layer. Root hairs that are found in the piliferous layers are always unicellular. They absorb water and mineral salts from the soil. Root hairs are generally short lived. The main function of piliferous layer is protection of the inner tissues.